

Firm's Labor Market Power:

A Review of Literature

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Abstract

Market power—the capacity of a firm to set price above marginal cost in the product market (a *markup*) or to pay a wage below the marginal revenue product of labour in the factor market (a *markdown*)—has returned to the centre of macroeconomics, industrial organisation and development economics. This review traces the intellectual history and the technical state of the art of measuring both faces of market power. We begin with the conceptual foundations (the Lerner index, the markup, and the markdown) and the welfare logic that makes these objects interesting. We then develop the *production approach*, from Hall (1988)'s reinterpretation of the Solow residual to the firm-level cost-minimisation estimator of De Loecker & Warzynski (2012). Because that estimator stands or falls on a credible estimate of the output elasticity of a flexible input, we give an extended treatment of the production-function toolkit—Olley & Pakes (1996), Levinsohn & Petrin (2003), Akerberg, Caves, & Frazer (2015), Wooldridge (2009) and the non-parametric identification of Gandhi, Navarro, & Rivers (2020)—of the choice between Cobb–Douglas and translog technology and between Hicks-neutral and factor-augmenting productivity (Demirer, 2025), and of the identification critiques that define the frontier: the revenue-elasticity problem (Bond et al., 2021), the internal-consistency test of Raval (2023), and the biases in markups read from financial statements (De Ridder, Grassi, & Morzenti, 2026; Fernald et al., 2026).

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We then turn to the factor side: the theory of monopsony from Robinson (1933) to Manning (2003) and the granular general-equilibrium model of Berger, Herkenhoff, & Mongey (2022); the three families of markdown measurement; and the striking recent finding that the secular rise in US market power is driven by *wage markdowns rather than price markups* (Ren & Zhang, 2025). We close with the empirical record for the world and for India and with the research gap the thesis fills.

Keywords: market power; markups; markdowns; monopsony; production function; output elasticity; De Loecker–Warzynski; Indian manufacturing.

JEL: D24, D43, J42, L11, L13, C13, O14.

1 Introduction: the two faces of market power

A firm exercises *market power* when it can profitably move a price away from the competitive benchmark in which price equals marginal cost. Two such departures organise the whole of this thesis. In the *product market*, a firm with pricing power charges a price P above its marginal cost of production c ; the ratio $\mu \equiv P/c$ is the *markup*. In the *factor market for labour*, a firm with wage-setting power pays a wage w below the marginal revenue product of labour MRPL; the ratio $\psi \equiv \text{MRPL}/w$ is the *markdown*. The two are conceptually symmetric—one is monopoly power over what the firm sells, the other is monopsony power over what the firm buys—and, as Sections 4 and 6 make clear, they are estimated with a single, unified econometric apparatus. They also have a common consequence that motivates the thesis: both drive a wedge between the value a worker produces and the income the worker receives, and so both depress the *labour share* of value added and distort the allocation of labour across firms.

The renewed interest in market power is easy to date. De Loecker, Eeckhout, & Unger (2020) show that the sales-weighted average markup of US-listed firms rose from about 1.21 in 1980 to about 1.61 by 2016, with the increase concentrated in the upper tail of the distribution and accompanied by a reallocation of sales toward high-markup firms. Autor et al. (2020) document that the fall in the US labour share is concentrated in industries that have become more concentrated—the “superstar firm” hypothesis—while Karabarbounis & Neiman (2014) establish that the labour-share decline is global, spanning more than fifty countries. On the factor side, Manning (2021) and the meta-analysis of Sokolova & Sørensen (2021) report that the elasticity of labour supply to the individual firm is typically only 1–4, far below the infinite value the competitive model assumes, so that wage markdowns of 20–50% are pervasive. The macroeconomic stakes—for the labour share, for inequality, for investment, for the transmission of monetary policy, and for aggregate productivity—are surveyed by Syverson (2019, 2025).

For a developing economy such as India the questions are sharper still. India entered the 1990s with a heavily regulated product market and a labour market segmented by low

geographic mobility, firm-specific amenities and search frictions—precisely the conditions under which the textbook assumption of wage-taking firms is least defensible. Yet there is almost no work that measures *both* the product-market and the factor-market power of Indian manufacturing plants, or that connects that power to the allocation of labour. This review supplies the conceptual and methodological foundation for doing so. Section 2 fixes definitions and the welfare accounting; Section 3 develops the production approach to the markup; Section 4—the technical core—reviews how the output elasticity is estimated and the identification problems that attend it; Sections 5–6 develop monopsony theory and the measurement of markdowns; Section 7 reviews the empirical record from the world to India; Section 8 sketches the determinants of market power (taken up fully in Review D); and Section 9 states the open debates and the gap the thesis fills.

2 Conceptual foundations and definitions

2.1 The Lerner index and the markup

Let a firm sell quantity Q at price P and let c denote its marginal cost. The *Lerner index* of monopoly power is

$$\mathcal{L} \equiv \frac{P - c}{P} = 1 - \frac{c}{P} = 1 - \frac{1}{\mu}, \quad \mu \equiv \frac{P}{c} \geq 1. \quad (1)$$

The index runs from 0 (perfect competition, $P = c$, $\mu = 1$) to 1 (the limiting monopoly) and is a monotone transformation of the markup. Its usefulness is that a monopolist's first-order condition links it to the price elasticity of demand $\varepsilon_D \equiv -\partial \ln Q / \partial \ln P > 0$:

$$\mathcal{L} = \frac{1}{\varepsilon_D} \quad \Longleftrightarrow \quad \mu = \frac{\varepsilon_D}{\varepsilon_D - 1}. \quad (2)$$

Equation (2) carries the central intuition of the demand-side tradition: market power is the inverse of the demand elasticity the firm faces. The more substitutable a firm's product (the larger ε_D), the closer price is forced to marginal cost. It also exposes the obstacle that motivated the modern literature: ε_D , c and even Q (as distinct from revenue PQ) are rarely observed, so markups cannot be read directly off (1)–(2).

2.2 The markdown and the symmetry of monopoly and monopsony

The factor-market analogue replaces the downward-sloping product demand curve with an upward-sloping labour-supply curve facing the firm, $L = L(w)$ with finite elasticity $\varepsilon_L \equiv d \ln L / d \ln w < \infty$. A wage-setting firm equates the marginal revenue product of labour to the *marginal cost of labour*, which exceeds the wage because hiring one more

worker raises the wage paid to all inframarginal workers:

$$\text{MCL} \equiv \frac{d(wL)}{dL} = w \left(1 + \frac{1}{\varepsilon_L} \right), \quad \text{MRPL} = \text{MCL} \implies \psi \equiv \frac{\text{MRPL}}{w} = 1 + \frac{1}{\varepsilon_L}. \quad (3)$$

The markdown ψ is the factor-market twin of the markup: as $\varepsilon_L \rightarrow \infty$ the firm becomes a wage-taker and $\psi \rightarrow 1$; as ε_L falls, the wedge between what a worker produces and what the worker is paid widens. This formulation is due to Robinson (1933), who coined “monopsony,” and is the static skeleton on which the dynamic models of Manning (2003) are built. Its empirical bite is large: the consensus $\varepsilon_L \approx 1.5$ of Sokolova & Sørensen (2021) implies $\psi \approx 1.67$, a wage roughly 40% below marginal revenue product.

2.3 A taxonomy of measurement strategies

Three broad routes lead to these objects, and the thesis uses the third. (i) *The demand/IO approach* estimates ε_D from an estimated demand system and recovers μ from (2); it is data-hungry and market-specific. (ii) *The accounting / price-cost-margin (PCM) approach* proxies the Lerner index by (sales – variable cost)/sales; it is simple but conflates markups with fixed-cost recovery and returns to scale. (iii) *The production approach* infers the markup (and the markdown) from the firm’s cost-minimising input choices and an estimate of technology, without specifying demand or the mode of competition. Because it needs only output, a flexible input, and that input’s expenditure, it is the workhorse for plant-level data such as the Indian Annual Survey of Industries (ASI). Its cost is that it places the whole identifying burden on a credible output elasticity—the subject of Section 4.

3 The production approach to markups: from Hall to De Loecker–Warzynski

3.1 Hall’s reinterpretation of the Solow residual

The modern production approach begins with Hall (1988), who showed that, once competition is not assumed, the Solow residual is contaminated by the markup. Consider a value-added technology $Q = \Omega F(N, K)$ with Hicks-neutral productivity Ω and a single variable input N . Cost minimisation with a competitive input market gives $w = c F_N$; multiplying by $N/(PQ)$ and using $\mu = P/c$ yields the accounting identity

$$\alpha_N \equiv \frac{wN}{PQ} = \frac{\theta_N}{\mu}, \quad \theta_N \equiv \frac{\partial \ln Q}{\partial \ln N}, \quad (4)$$

so a variable input's *revenue* share understates its output elasticity by exactly the factor μ . Under constant returns ($\theta_N + \theta_K = 1$), differencing gives Hall's estimating equation

$$\Delta q_t - \Delta k_t = \mu \alpha_N (\Delta n_t - \Delta k_t) + \Delta \omega_t, \quad (5)$$

whose slope identifies $\mu \alpha_N$; dividing by the observed labour share gives μ . Because $\Delta \omega_t$ is correlated with input growth, Hall (1988) instruments with demand shifters. Two features survive into all later work: the markup is the wedge between an output elasticity and a revenue share, and identifying it means separating that wedge from productivity.

3.2 The De Loecker–Warzynski estimator

De Loecker & Warzynski (2012) recast Hall's insight at the level of the individual firm and a single cross-section, dispensing with constant returns, with a competitive output market, and with industry aggregation. Let firm i at time t have gross-output technology $Q_{it} = F(\mathbf{x}_{it}) e^{\omega_{it}}$ in inputs $\mathbf{x} = (K, L, M)$, and let V be a *variable* input chosen statically, without adjustment costs, from a market in which the firm is a price-taker (paying P_{it}^V). Cost minimisation makes the Lagrange multiplier on the output constraint equal to marginal cost, $\lambda_{it} = c_{it}$. The first-order condition for V , $P_{it}^V = \lambda_{it} \partial Q_{it} / \partial V_{it}$, multiplied by V_{it} / Q_{it} and divided by $P_{it} = \mu_{it} \lambda_{it}$, gives the central formula of the production approach,

$$\boxed{\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V}, \quad \theta_{it}^V \equiv \frac{\partial \ln Q_{it}}{\partial \ln V_{it}}, \quad \alpha_{it}^V \equiv \frac{P_{it}^V V_{it}}{P_{it} Q_{it}}.} \quad (6)$$

The economic content is worth stating slowly. The numerator θ_{it}^V is a *technological* object—how much output rises, in percentage terms, when the flexible input rises by one percent—and is estimated from the production function. The denominator α_{it}^V is an *accounting* object, the input's expenditure share in sales, read from the data. If the firm were a price-taker in the output market ($\mu = 1$), cost minimisation would force the elasticity to equal the revenue share and the ratio would be one. Whenever the firm prices above marginal cost it optimally restricts output and uses less of the flexible input than the revenue share would imply, so the elasticity exceeds the share and the ratio recovers the markup. Crucially, (6) invokes *no* assumption about demand, the number of competitors, or strategic conduct: it follows from cost minimisation alone. Its requirements are exactly three—a genuinely flexible input, price-taking in that input's market, and a credible estimate of θ^V —and the credibility of the third is the content of Section 4.

3.3 From one flexible input to the labour markdown

The same logic, applied to a second input, delivers the markdown and motivates Reviews B and C. Suppose materials M are flexible and bought competitively (no monopsony in materials), while labour L may be subject to wage-setting power. Applying (6) to each input and taking the ratio, the demand-side markup μ cancels and what remains is the labour wedge

$$\psi_{it} = \frac{\mu_{it}^L}{\mu_{it}^M} = \frac{\theta_{it}^L / \alpha_{it}^L}{\theta_{it}^M / \alpha_{it}^M}, \quad (7)$$

which is exactly the Brooks et al. (2021) markdown and the object the thesis estimates plant-by-plant; $\psi_{it} > 1$ signals a wage below marginal revenue product. The identity uses a competitively-priced flexible input (materials) as an internal numeraire that nets out product-market power, isolating factor-market power. It also makes transparent why everything depends on the output elasticities θ^L and θ^M : every markup and markdown in the thesis is a ratio whose numerator must be estimated. We turn to that estimation.

4 Estimating the output elasticity: the production-function toolkit

This section is the technical heart of the review. The markup formula (6) and the markdown formula (7) are only as good as the estimated output elasticities they contain. Estimating a production function consistently is one of the oldest hard problems in econometrics, because the inputs a firm chooses are correlated with productivity the analyst does not observe. We state the two classical identification problems, trace the proxy-variable solutions (OP, LP, ACF, Wooldridge), present the non-parametric identification of Gandhi, Navarro, & Rivers (2020), discuss functional form and the productivity concept, and end with the critiques that bound what the production approach can credibly claim.

4.1 Two classical identification problems

Write the log gross-output production function as

$$q_{it} = f(\mathbf{x}_{it}; \boldsymbol{\beta}) + \omega_{it} + \varepsilon_{it}, \quad (8)$$

where $\mathbf{x}_{it} = (k_{it}, \ell_{it}, m_{it})$ are log inputs, ω_{it} is productivity known to the firm when it chooses inputs, and ε_{it} is an *ex post* shock the firm does not anticipate.

(1) *Simultaneity / transmission bias* (Marschak & Andrews, 1944). Because ω_{it} is seen by the firm but not the analyst, and more productive firms use more inputs, the regressors in (8) are correlated with the omitted ω_{it} ; OLS is biased—up on flexible inputs,

down on capital. A consistent estimator must control for ω_{it} .

(2) *Output-price / revenue bias*. Surveys record *revenue* $R_{it} = P_{it}Q_{it}$, not physical quantity. Deflating by an industry price index leaves unobserved firm-specific price variation in the residual. Under imperfect competition this is not innocuous: as Section 4.8 shows, the elasticity estimated from deflated revenue is a *revenue* elasticity, not the output elasticity (6) requires (Bond et al., 2021).

4.2 Olley–Pakes: investment as a proxy for productivity

Olley & Pakes (1996) solve simultaneity with a control function. In a dynamic investment model optimal investment is strictly increasing in productivity given capital, $i_{it} = i(\omega_{it}, k_{it})$, so it can be inverted: $\omega_{it} = h(i_{it}, k_{it})$. Substituting into (8) gives a partially linear model. A first stage regresses output on the variable input(s) and a non-parametric $\phi(i_{it}, k_{it}) = \beta_k k_{it} + h(i_{it}, k_{it})$, identifying the variable-input coefficient(s) and purging ε_{it} . A second stage assumes a first-order Markov productivity process $\omega_{it} = g(\omega_{it-1}) + \xi_{it}$ with ξ_{it} orthogonal to lagged information, and identifies β_k from $\mathbb{E}[\xi_{it} k_{it}] = 0$ (capital is predetermined). Olley & Pakes (1996) also correct for endogenous *exit*. The limitation is that investment is lumpy and often zero, so the monotonicity fails at zero and many observations are discarded.

4.3 Levinsohn–Petrin: intermediate inputs as the proxy

Levinsohn & Petrin (2003) keep the control-function logic but use an intermediate input—materials—as the proxy, inverting $\omega_{it} = h(m_{it}, k_{it})$. Because materials are rarely zero and adjust smoothly, far fewer observations are lost and monotonicity is more plausible. This is the estimator most natural for the ASI, where materials are well measured but firm-level investment is noisy; it is the proxy in the thesis’s pipeline.

4.4 Akerberg–Caves–Frazer: the functional-dependence critique and its fix

Akerberg, Caves, & Frazer (2015) identify a flaw in the OP/LP first stage. If labour ℓ_{it} is chosen at the same time as the proxy and as a function of the same states (ω_{it}, k_{it}) , then ℓ_{it} is itself a function of (ω_{it}, k_{it}) ; but the first-stage control function already conditions non-parametrically on the proxy and capital, which span the same information, so labour is *collinear* with the non-parametric term and its coefficient is not separately identified—the “functional-dependence problem.” The ACF remedy restructures the moments: the first stage only nets out ε_{it} and recovers expected output $\Phi_{it} = f(\mathbf{x}_{it}; \beta) + \omega_{it}$, claiming *no* input coefficient; all coefficients are identified in a second stage by GMM, forming

$$\xi_{it}(\beta) = \omega_{it}(\beta) - \mathbb{E}[\omega_{it}(\beta) \mid \omega_{it-1}(\beta)], \quad \omega_{it}(\beta) = \Phi_{it} - f(\mathbf{x}_{it}; \beta), \quad (9)$$

and using $\mathbb{E}[\xi_{it}(\boldsymbol{\beta}) \mathbf{z}_{it}] = 0$. The instruments $\mathbf{z}_{it}$ follow from the *timing* of input decisions: capital is predetermined and instruments for itself; labour, if chosen at $t - 1$ or an intermediate sub-period, instruments for itself through its lag; lagged flexible inputs add moments. The identifying assumptions are thus made explicit as assumptions about *when* each input is chosen. ACF is the default in applied work and the estimator behind the thesis’s markups; it still requires a credible monotone proxy and the timing assumptions.

4.5 Wooldridge’s one-step GMM and software

Wooldridge (2009) combines the two stages into one GMM system stacking the ε - and ξ -moments: more efficient, with robust standard errors and no ad hoc bootstrap, and immune to the functional-dependence problem by construction. Stata’s `prodest` (ACF and Wooldridge options) and R implementations follow this template (Rovigatti & Mollisi, 2018). These refinements do not change the economics but materially affect the precision of the elasticities that feed (6).

4.6 Gandhi–Navarro–Rivers: non-parametric identification from the share equation

A deeper question is whether the gross-output production function is identified at all by the proxy approach. Gandhi, Navarro, & Rivers (2020) show that, without more structure, it is not: the flexible-input elasticity cannot be recovered from the output equation alone. Their solution uses the information proxy methods discard—the firm’s first-order condition for the flexible input. For a flexible input M bought competitively at price ρ_{it} , profit maximisation under the ex post shock implies $\rho_{it}M_{it} = P_{it}Q_{it}\theta^M(\mathbf{x}_{it})/\mathcal{E}$ with $\mathcal{E} \equiv \mathbb{E}[e^\varepsilon]$; in logs this is the *share equation*

$$s_{it} \equiv \ln \frac{\rho_{it}M_{it}}{P_{it}Q_{it}} = \ln \theta^M(\mathbf{x}_{it}) - \ln \mathcal{E} - \varepsilon_{it}. \quad (10)$$

Since the left side is observed and the right side is a known transformation of the inputs plus the ex post shock, a non-parametric regression of the log revenue share on $\mathbf{x}_{it}$ identifies the flexible-input elasticity (and $\mathcal{E}$) *without* inverting for productivity. Integrating θ^M/M over M recovers the production function up to a “constant of integration” $\mathcal{C}(k, \ell)$, pinned down from the productivity law of motion via $\mathbb{E}[\xi_{it} \mathbf{z}_{it}] = 0$. Two implications matter. First, (10) *is* the cost-minimisation condition behind (6), so the GNR programme places markup estimation on a rigorous non-parametric footing for the flexible input. Second, the markup-relevant elasticity comes from input-share data, which is reassuring—though it does not by itself resolve the price/quantity problem of Section 4.8.

4.7 Functional form and the concept of productivity

Cobb–Douglas versus translog. Cobb–Douglas (CD) imposes constant elasticities $\theta_{it}^V = \beta_V$, so all markup variation in (6) comes from revenue shares. Translog lets elasticities vary with the input mix, $\theta_{it}^V = \beta_V + 2\beta_{VV}v_{it} + \sum_{X \neq V} \beta_{VX}x_{it}$, admitting heterogeneous elasticities and richer markup heterogeneity, at the cost of many parameters and a heavy demand on within-firm variation. In short rotating panels such as the ASI, where the median plant appears in one or two waves, translog is often unidentified or yields implausible returns to scale; this is why the thesis adopts CD, a choice shared by Brooks et al. (2021) on comparable data.

Hicks-neutral versus factor-augmenting technology. The standard estimators model productivity as Hicks-neutral—a single ω_{it} that scales all inputs equally. Much applied work also imposes parametric (e.g. CES) forms (De Loecker & Syverson, 2021). Demirer (2025) shows this is not innocuous for markups: when technical change is in fact factor-augmenting (e.g. labour-augmenting), Hicks-neutral parametric restrictions mismeasure output elasticities and their cross-firm heterogeneity—and, because biased flexible-input elasticities translate one-for-one into spurious markups (Raval, 2023), can even *reverse* the apparent time trend in markups. His non-parametric estimator, applied to US and four developing-country manufacturing samples, shows that whether markups appear to rise or fall can hinge on the assumed direction of technical change. For a thesis that reports a *falling* mean markup in Indian manufacturing, this is a robustness frontier worth explicit attention.

4.8 Identification critiques and the current frontier

The production approach is mature enough to have generated a sharp critical literature.

The revenue-elasticity problem. Bond et al. (2021) make the most fundamental objection. With only deflated revenue, one estimates the elasticity of *revenue*, not of *output*. Under a downward-sloping demand curve revenue responds less than output to an input increase, so the revenue elasticity equals the output elasticity scaled by the inverse markup, $\theta^{V,\text{rev}} = \theta^V / \mu$; substituting into the markup formula returns $\theta^{V,\text{rev}} / \alpha^V = 1$, an uninformative unit markup. Absent physical-quantity or price data or extra demand structure, the markup *level* is not identified from revenue alone, and estimated markup *variation* must come from revenue-share variation whose interpretation is delicate.

What survives: dispersion and trends. De Ridder, Grassi, & Morzenti (2026) sharpen this into constructive guidance. Because financial statements report a firm’s revenue but not the physical quantity it produces, the estimated variable-input elasticity is biased and the *average* revenue-based markup is uninformative about the true average. But—and this is the rehabilitating result—the *dispersion* of markups across firms and their *trend* over time can be recovered well from financial data *provided firms share*

a common production technology (as under Cobb–Douglas, and more broadly under empirically relevant conditions): the bias enters as a multiplicative factor of unknown direction that preserves the relative ordering and the sign of time trends. They validate this on French firm-level production and pricing data. The practical lesson is to trust production-approach *patterns* far more than production-approach *levels*.

Internal-consistency tests. Raval (2023) proposes a direct check: the approach should give the same markup whether computed from the labour or the materials first-order condition, since both equal P/c . Empirically the two input-based markups are often *negatively* correlated across firms—impossible if the model is correctly specified—which points to mismeasured output elasticities. The implication is to triangulate across inputs rather than trust a single flexible factor.

The markup is a residual. Fernald et al. (2026) frame the deepest cautionary point: the production-based markup is fundamentally a residual, and like the Solow residual it is clean in theory but in practice contaminated by misspecification and mismeasurement, so that small differences in implementation—which input is treated as flexible, how overhead and fixed costs are handled, how the user cost of capital is measured—can produce starkly different answers from the very same data. This, they argue, is why the literature disagrees about whether US markups rose sharply or barely at all, and they call for far greater transparency about implementation choices. Relatedly, Ruzic & Ho (2021) show in US Census manufacturing data that allowing for industry-specific returns to scale and markups can turn a measured *increase* in misallocation under the usual constant-markup, constant-returns assumptions into a measured *decrease*—a result that connects directly to Reviews B and C. The lesson for the thesis is methodological humility: plant-level levels are less trustworthy than patterns, robustness to the productivity concept and to returns to scale is essential, and markup and markdown estimates should be read as carefully bounded objects.

4.9 Summary of the toolkit

Table 1 collects the estimators and their identifying content. The thread is unbroken from Hall (1988) to the present: the markup is an elasticity-to-share wedge; the elasticity must be estimated while controlling for an unobserved productivity that the firm sees and the analyst does not; and the credibility of any markup or markdown is exactly the credibility of that elasticity. We now turn to the factor side—monopsony, the markdown, and the empirical record—building directly on (6)–(7).

5 Labour-market power: the theory of monopsony

Estimator	Key device	Identifying content / limitation
Hall (1988)	Solow residual + demand IV	Markup as elasticity/share wedge; needs CRS and aggregate demand instruments.
Olley–Pakes (1996)	Investment proxy for ω ; exit correction	Solves simultaneity and selection; drops zero-investment plants.
Levinsohn–Petrin (2003)	Materials proxy for ω	Less data loss; smoother monotonicity; natural for ASI.
Akerberg–Caves–Frazer (2015)	Two-stage GMM with timing moments	Fixes functional-dependence; identifies all coefficients in stage 2; the thesis’s choice.
Wooldridge (2009)	One-step GMM	Efficient, robust SEs; avoids collinearity by construction.
Gandhi–Navarro–Rivers (2020)	Flexible-input share equation	Non-parametric identification of the flexible elasticity; grounds DLW.
Demirer (2025)	Factor-augmenting, non-parametric	Hicks-neutrality can mismeasure or reverse markup trends.
<i>Critiques:</i> Bond et al. (2021) revenue-elasticity (level not identified); De Ridder et al. (2026) financial-data biases (dispersion/trends survive under common technology); Raval (2023) cross-input inconsistency; Fernald et al. (2026) the markup as a contaminated residual; Ruzic (2021) returns-to-scale and the misallocation trend.		

Table 1: The production-function toolkit for estimating output elasticities and markups.

5.1 Static monopsony and the source of the markdown

The whole substance of (3) lies in *why* a firm faces a finite labour-supply elasticity ε_L . Robinson (1933) introduced the static picture: a firm that wants more labour must post a higher wage, and because it pays that wage to everyone, the marginal cost of labour $MCL = w(1 + 1/\varepsilon_L)$ lies above the wage. Profit maximisation sets $MRPL = MCL$, leaving a positive markdown $\psi = 1 + 1/\varepsilon_L$ and a wage strictly below marginal revenue product. The model is silent about the source of the upward slope; the modern literature supplies it. Workers are not perfectly mobile: search is costly, information is incomplete, jobs differ in non-wage amenities, and—especially in India—geographic and social mobility is low. Each of these gives the individual employer a slice of wage-setting power even in a market with many employers, so monopsony is better thought of as pervasive “oligopsony” than as the textbook single buyer.

5.2 Dynamic monopsony: search foundations and the Manning identity

Manning (2003), building on the equilibrium search model of Burdett & Mortensen (1998), turns the static curve into a dynamic one. A firm’s workforce is a stock fed by recruitment and drained by separations: in steady state $s(w)L = R(w)$, where the

separation rate $s(w)$ falls and recruitment $R(w)$ rises with the wage. Differentiating, the firm’s labour-supply elasticity decomposes into a separation and a recruitment margin,

$$\varepsilon_L = \varepsilon_R(w) - \varepsilon_s(w), \quad \text{and, under the search symmetry of Burdett–Mortensen,} \quad \varepsilon_L \approx -2\varepsilon_s(w), \quad (11)$$

so the elasticity that governs the markdown can be *estimated from labour flows*: how strongly quits respond to the wage tells us how much wage-setting power the firm has. The empirical payoff of (11) is that it converts an unobservable (the slope of the firm’s labour-supply curve) into observables (separation and recruitment elasticities), and it is the conceptual bridge between the production-approach markdown of (7) and the direct, flow-based estimates discussed in Section 6.1.

5.3 General-equilibrium and granular models

The frontier embeds monopsony in general equilibrium. Berger, Herkenhoff, & Mongey (2022) build a model of granular oligopsony in local labour markets: a finite number of firms compete in wages within nested-CES labour markets, each internalising its effect on the local wage, so that the equilibrium markdown rises with a firm’s local employment share and with labour-market concentration. Calibrated to US data, the model implies that wages fall substantially below the competitive level even with many employers, and that the welfare losses from labour-market power are first-order. A complementary, more reduced-form tradition—Card et al. (2018) on rent-sharing and firm wage premia and Lamadon, Mogstad, & Setzler (2022) on imperfect competition and compensating differentials—reaches the same destination from the wage side: firms pay systematically different wages for identical workers, and a meaningful part of the wage distribution reflects employer power rather than worker heterogeneity. Together these models establish that the markdown is not a curiosity of a few company towns but a structural feature of labour markets, with direct implications for the labour share and for the allocation of workers across firms that Reviews B and C take up.

6 Measuring labour-market power

Three families of method dominate; the thesis uses the third, and the value of reviewing all three is that they discipline one another.

6.1 Direct labour-supply-elasticity estimates

The most transparent route estimates ε_L directly. Following (11), one regresses separation (and recruitment) rates on the firm wage, or exploits quasi-experimental wage variation (minimum-wage changes, mergers, firm-specific shocks) to trace out the firm’s labour-

supply curve. The accumulated evidence is decisive: the meta-analysis of Sokolova & Sørensen (2021), pooling 67 studies, finds an average firm-level labour-supply elasticity of about 1.5, implying by (3) a wage roughly 40% below marginal revenue product; Manning (2021) summarises the range as 1–4. These numbers are an order of magnitude away from the competitive benchmark of $\varepsilon_L = \infty$.

6.2 Concentration-based measures

A second family infers power from market structure, computing a labour-market Herfindahl index over employers within a commuting zone and occupation and relating it to wages. Azar, Marinescu, & Steinbaum (2022) document that US labour markets are often highly concentrated and that higher concentration is associated with lower posted wages. The approach is data-light and intuitive but rests on a contestable definition of the relevant labour market and treats concentration as exogenous when it is itself an equilibrium outcome; it is therefore best read as corroborating, not replacing, structural estimates.

6.3 The production approach to the markdown

The third family is the one used in this thesis and is the natural extension of Section 3. From (7), with materials as a competitively-priced flexible numeraire, the markdown on labour is $\psi = \mu^L/\mu^M = (\theta^L/\alpha^L)/(\theta^M/\alpha^M)$. The identifying assumption is explicit and consequential: the numeraire input must be flexible and free of monopsony. Yeh, Macaluso, & Hershbein (2022) implement this on US Census manufacturing microdata (1976–2014) and find that most plants operate in a monopsonistic environment, with an average markdown of about 1.53—a worker earning roughly 65 cents on the marginal dollar of value created—and that the aggregate markdown fell from the late 1970s to the early 2000s before rising sharply thereafter. Mertens (2022) reaches compatible conclusions for Germany, linking rising markdowns to the fall in the labour share.

The most striking recent contribution is Ren & Zhang (2025), who decompose firm-level market power into a product component (the markup $\mu = P/\text{MC}$) and a labour component (the markdown $\psi = \text{MRPL}/w$) using matched US Census and Compustat data, 1977–2019. They document three facts that bear directly on this thesis: (i) the secular rise in aggregate US market power is driven by *wage markdowns*, while *price markups have remained essentially stable*; (ii) larger and more productive firms have *lower* product-market power but *higher* labour-market power, so the familiar size–markup correlation partly reflects monopsony; and (iii) the widely observed incomplete pass-through of cost shocks into prices reflects monopsony power rather than the curvature of demand. Their reconciliation with De Loecker, Eeckhout, & Unger (2020) is illuminating and methodologically important: when a single labour-inclusive flexible input (such as the cost of goods sold) is used to estimate the markup, the recovered object is not the

markup but the product of markup and markdown,

$$\ln \mu_{it}^{\text{measured}} = \ln \mu_{it} + \ln \psi_{it}, \quad (12)$$

so that what earlier work read as “rising markups” is, once labour is separated from the flexible input, largely *rising markdowns*. Equation (12) is a caution every applied study—including this thesis—must heed: the choice of flexible input determines whether one is measuring product power, factor power, or an unidentified blend of the two. It is precisely because the thesis uses *materials* (not a labour-inclusive composite) as the numeraire that (7) can separate μ^M from ψ .

6.4 What the families share, and a caution

All three families converge on a single qualitative conclusion—firm-level labour-supply elasticities are low and wage markdowns are pervasive—but the production approach inherits the identification problems of Section 4. The revenue-elasticity critique (Bond et al., 2021), the cross-input inconsistency of Raval (2023), and the warning of Hashemi, Kirov, & Traina (2022) that the production approach can mistake genuine input distortions (adjustment costs, regulatory wedges) for markdowns all apply to ψ exactly as they apply to μ . The disciplined response, and the one this thesis adopts, is to anchor levels cautiously, to report distributions and trends rather than rely on point levels, and to read the production-approach markdown alongside the direct and concentration-based evidence.

7 The empirical record: from the world to India

7.1 Markups: levels, trends, and the caveats

The headline fact is the apparent rise of US markups in De Loecker, Eeckhout, & Unger (2020), from about 1.21 in 1980 to 1.61 in 2016, concentrated in the upper tail and accompanied by a reallocation of sales to high-markup firms. As Section 4.8 stressed, this trend is now contested on three fronts: the levels may be biased by the revenue-elasticity problem (Bond et al., 2021); the magnitude is sensitive to implementation choices that make the markup a contaminated residual (Fernald et al., 2026); and—most consequentially for this thesis—Ren & Zhang (2025) show that, once labour is separated from the flexible input, the aggregate *markup* is roughly flat and the rise in total market power is a rise in *markdowns*. The reasonable reading is that there is robust evidence of rising market power broadly defined, considerable uncertainty about its split between product and labour markets, and strong reason to measure both rather than markups alone.

7.2 Markdowns and the labour share

The factor side is where the evidence has hardened. Direct elasticity estimates (Sokolova & Sørensen, 2021; Manning, 2021), production-approach markdowns (Yeh, Macaluso, & Hershbein, 2022; Mertens, 2022; Ren & Zhang, 2025), and the granular general-equilibrium model of Berger, Herkenhoff, & Mongey (2022) agree that wage markdowns are large and, in recent decades, rising. This dovetails with the macro literature on the labour share: the global decline documented by Karabarbounis & Neiman (2014), its concentration in concentrating “superstar” industries (Autor et al., 2020), and the simultaneous fall in both labour and capital shares with a rising profit residual (Barkai, 2020). Market power—product, factor, or both—is the common thread.

7.3 India

The Indian evidence is thinner and, tellingly, almost entirely about the product market. A first generation of studies measured price–cost margins and their response to the 1991 reforms: Balakrishnan et al. (2006) examine liberalisation, market power and productivity growth in Indian industry, and Goldberg et al. (2010) show that access to imported intermediate inputs raised product scope and, by lowering input costs, plausibly compressed markups. More recent work using the production approach—Mukherjee & Chanda (2021) on tariff liberalisation and firm-level markups—finds the markup–trade relationship to be two-sided and sensitive to whether output or input tariffs move. What is almost entirely missing is the *factor* side: there is very little estimation of labour-market power for Indian manufacturing, and essentially none that measures product and factor power jointly on the same plants. India is also a setting where the monopsony mechanisms of Section 5 are unusually strong—low geographic mobility, a large informal margin, and protective labour regulation (notably the Industrial Disputes Act and the rise of contract labour) that segments the workforce. The present thesis is built to fill this gap. Using the ASI plant panel for 2012-13 to 2023-24, it finds a mean product markup that *falls* from about 1.88 to 1.73 (a decline of roughly 8%, against the global upward narrative), a mean labour markdown well above one throughout (with more than two-thirds of plants paying below marginal revenue product), an implied firm-level labour-supply elasticity near 0.3 (consistent with the low-mobility prior), and a labour share that falls from about 22% to 16% of value added. In the language of (12), the Indian case looks less like a product-market story and more like a factor-market one—exactly the hypothesis Reviews B and C develop.

8 Determinants of market power

What gives rise to market power in the first place? The literature points to several forces, each treated in depth in Review D and noted here for completeness. *Technology and “superstar” dynamics*: scalable, intangible-intensive technologies let the most productive firms capture rising market share, raising concentration and, in the product market, markups (Autor et al., 2020); on the labour side Ren & Zhang (2025) show the same superstar dynamics raise *markdowns*, because larger firms face less elastic labour supply. *Trade openness*: output-tariff reductions discipline domestic markups while input-tariff reductions can raise them by lowering costs, a genuinely two-sided channel (De Loecker et al., 2016; Mukherjee & Chanda, 2021); in the labour market, import competition and export access reshape the elasticities that govern markdowns. *Concentration, entry barriers and regulation*: product- and labour-market concentration, licensing and employment-protection law all bear on the wedges. Because trade and macroeconomic determinants are central to Objectives 2 and 3 of the thesis, their full treatment is deferred to Review D; the point to carry forward is that markups and markdowns are equilibrium outcomes of these forces, not primitives.

9 Debates, open questions, and the research gap

Four debates frame the current literature and locate the thesis within it. *Levels versus patterns*: production-approach *levels* of markups and markdowns are fragile—unidentified from revenue alone (Bond et al., 2021), sensitive to implementation (Fernald et al., 2026)—while *dispersion and trends* are far more robust, surviving under a common production technology (De Ridder, Grassi, & Morzenti, 2026). *Markups versus markdowns*: the reinterpretation of Ren & Zhang (2025), that rising “market power” is largely rising monopsony once labour is separated from the flexible input, makes joint measurement of both wedges indispensable rather than optional. *Technology and functional form*: factor-augmenting productivity and returns to scale can reverse measured trends in both markups and misallocation (Demirer, 2025; Ruzic & Ho, 2021), demanding explicit robustness. *Distortions versus power*: the production approach can confound genuine factor-market distortions with markdowns (Hashemi, Kirov, & Traina, 2022), a point that becomes central in Review C, where market power and misallocation must be told apart.

Against this backdrop the gap the thesis fills is specific and, in the Indian context, almost unoccupied. First, the Indian literature measures product-market power in isolation and rarely, if ever, the factor-market power of the same plants; the thesis estimates *both* the markup μ and the markdown ψ on a single large ASI panel, using materials as a competitive numeraire to keep (12) from blending the two. Second, it does

so over a recent and policy-relevant window (2012-13 to 2023-24) that spans the Goods and Services Tax and the post-reform maturation of Indian manufacturing. Third—and this is the bridge to the remaining reviews—it connects these wedges to the *allocation* of labour: if markdowns are large and dispersed across plants, they are not only a distributional fact about the labour share but a source of *misallocation*, because identical workers command different marginal products at different employers. Review B develops the misallocation framework that makes this precise; Review C joins the two literatures; and Review D embeds them in the trade and macroeconomic determinants that Objectives 2 and 3 examine.

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